

3.1.4 Forming, redistribution and addition processes

Paper and board forming processes

Content	Potential links to maths and science
Students should be aware of the ways that paper and board can be shaped into different products such as packaging. Specific process to include: • die cutting • laser cutting • creasing • bending.	

Polymer processes

Content	Potential links to maths and science
Students should be aware of how polymers can be formed into 3D products. They should be able to describe the different forming methods.	
They should be able to explain the suitability of the different forming methods for a range of specific products and scales of production. Specific process to include: • vacuum forming • thermoforming • calendaring • line bending • laminating (layup) • injection moulding • blow moulding • rotational moulding • extrusion • compression moulding.	

Metal processes

Content	Potential links to maths and science
Students should be aware of how metals can be shaped into 3D products. They should be able to describe the different forming methods. They should be able to explain the suitability of the different forming methods for a range of specific products and scales of production. Specific processes to include: • press forming • spinning • cupping • deep drawing • forging • drop forging • bending • rolling • casting: • sand casting • die casting • investment casting • low temperature casting (pewter).	
Students should be aware of the different permanent and temporary joining methods for metals. They should be able to describe the different methods. They should be able to explain the suitability of the different joining methods for a range of specific products and scales of production. Including addition/fabrication processes: • metal inert gas (MIG) welding • tungsten inert gas (TIG) welding • spot welding • oxy-acetylene welding • soldering (soft and hard) • brazing • riveting • temporary joining methods and fasteners: • self tapping screws • machine screws • nuts and bolts.	

Content	Potential links to maths and science
Students should be aware of the different wasting processes. They should be able to describe the different processes. They should be able to explain the suitability of the different wasting processes for a range of specific components and products. Specific processes to include: • milling • turning • flame cutting • plasma cutting • laser cutting • punching/stamping.	

Wood processes

Content	Potential links to maths and science
Students should be aware of how timber can be joined to form different products. They should be able to describe the different methods. They should be able to explain the suitability of the different joining methods for a range of specific products and scales of production. Including: • addition/fabrication processes • traditional wood jointing: • dovetail joint • comb joint • housing joint • half-lap joint • dowel joint • mortise and tenon • component jointing: • knock down (KD) fittings • wood screws • nuts and bolts • coach bolts.	

Content	Potential links to maths and science
Students should be aware of how timber can be formed into 3D products. They should be able to describe the different processes. They should be able to explain the suitability of the different wasting processes for a range of specific products. Specific processes to include: • laminating • steam bending • machine processes: • turning between centre • use of the chuck and faceplate • milling • routing to produce slots, holes and profiles.	

3.1.4.5 The use of adhesives and fixings

Content	Potential links to maths and science
• PVA • Contact adhesives • UV hardening adhesive • Solvent cements such as Tensol or acrylic cement • Epoxy resin	

Jigs and fixtures

Content	Potential links to maths and science
Students should be aware of how jigs and fixtures can be used to aid the manufacture of products. They should be able to describe them and explain their suitability for accurate and repeated manufacture of products.	Dimensions and angles in the design of jigs, fixtures and templates.